Venkata Lakshmi Parimala Pasupuleti

Staff Software Engineer · Agentic AI Systems · Autonomous Pipelines · Human-in-the-Loop Controls · Scalable Architecture · Cross-Functional Technical Leadership

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PROFESSIONAL SUMMARY

Software engineer with 9+ years of production systems experience and a focus on building agentic AI platforms that advance from AI-assisted to autonomous operation - with the human-in-the-loop controls and regression gates that make autonomous systems safe to run at scale. LeaseI is the proof point: an 8-agent LangGraph platform I architected and shipped solo, with deterministic HITL escalation, staged autonomy across three surfaces, and a CI-integrated evaluation gate that blocked zero regressions across 1,240+ test cases. AgentProbe extends this pattern: an open-source LLM evaluation platform with multi-provider routing, 5-metric quality engine, and a CI regression gate that stops unsafe outputs before they ship. At Entrata, I built and owned high-stakes automated pipelines for 9+ years - systems where correctness, reliability, and auditability were non-negotiable (100K+ housing units, federal compliance). I thrive in cross-functional environments, have mentored 5+ engineers and 600+ students, and bring both the technical depth and the communication skills to drive alignment across engineering, product, and design. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

TECHNICAL SKILLS

Agentic AI & Autonomous Systems: LangGraph, LangChain, Multi-Agent Orchestration, Agentic Pipelines, Human-in-the-Loop (HITL), Guardrails, Tool Calling, Autonomous Routing, Staged Rollout, LLM Evaluation, LLM-as-Judge, Hallucination Detection, RAG Pipelines, Prompt Engineering, Anthropic API, OpenAI API, MCP, Claude Code

Systems & Infrastructure: Python (9yr), FastAPI, REST APIs, PostgreSQL, MySQL, Oracle, Redis, RabbitMQ, Docker, AWS (S3, EC2, RDS, Glue), CI/CD (GitHub Actions), Microservices, Distributed Systems, Multi-Tenant SaaS, System Design, Scalable Architecture, Linux, Git

Reliability & Cross-Functional Delivery: CI-Integrated Regression Gates, Testing Strategy (Unit, Integration, End-to-End), Observability & Monitoring, Production Debugging, Performance Engineering, Technical Leadership, Mentoring, Cross-Functional Stakeholder Delivery, Agile/Scrum, TypeScript, React.js, JavaScript

EXPERIENCE

Magizhs Tech Solution

May 2025 - Aug 2025

AI/ML Engineer - Agentic System Delivery

- Architected and shipped a production-grade agentic AI system (LangGraph, FastAPI, Python) from ambiguous client requirements to deployment - designed the AI-assisted → autonomous progression with staged rollout and HITL controls; owned architecture, testing, observability, and stakeholder handoff end-to-end using Claude Code to compress the build cycle; delivered in weeks with full autonomy.

George Mason University

Aug 2024 - May 2026

Graduate Teaching Assistant & Research Associate

- Built ML pipelines (PyTorch, scikit-learn) on WHO and UK Road Safety datasets - $R^2 = 0.977$; designed observability dashboards improving research comprehension 40%; mentored 600+ students in Python, SQL, and system design.

Entrata - Apartment & Affordable Housing SaaS

Sep 2018 - Aug 2024

Senior Software Engineer · Lehi, UT (Remote)

- **9+ years owning high-stakes automated pipelines where correctness, reliability, and auditability were non-negotiable** - owned the HUD income-limit compliance pipeline for 7 years (federal data → 180,000 domain records powering eligibility decisions for 100K+ housing units); designed scalable architecture from scratch and iterated on it in production across 120+ Agile sprints; 20+ customer-facing production modules shipped with full autonomy - no playbook, full ownership.
- **Infrastructure built to operate autonomously at scale** - RabbitMQ async processing with retry logic, idempotency guarantees, and dead-letter handling; Docker microservices (50% scalability improvement, 30% deployment time reduction); AWS S3/EC2/RDS/Glue; Redis caching; 200+ SQL query optimizations (30% performance gain); 99.5% uptime; 100% PHI compliance with zero security incidents over 7yr; built internal debugging tools reducing triage time by 40%.
- **Technical leadership and cross-functional delivery** - GitHub Actions CI/CD; maintained sub-10 bug count 6+ consecutive months; 300+ peer code reviews enforcing production-grade standards; mentored 5+ engineers; leveraged Claude Code and AI-assisted development to accelerate iteration; worked directly with product, compliance, and operations stakeholders across every sprint.

Entrata India - Pune

Jan 2017 - Sep 2018

Associate Software Engineer → Trainee

- Built backend modules (PHP, Python, PostgreSQL) for apartment management compliance; data pipeline for 1.6M HUD/Tax Credit records; AWS S3 migration resolving 90% of legacy storage constraints.

PROJECTS

Leasel: Agentic AI Platform - AI-Assisted → Agentic → Autonomous with HITL Controls

Agentic · HITL · Staged Autonomy · End-to-End

Python · LangGraph (8-agent) · FastAPI · TF-IDF · OCR · SQLite · Claude Code | github.com/venkatalakshmiparimala/Lease_complaince_rag

Three-surface production-grade agentic AI platform built solo from concept to deployment - advancing the autonomy curve: document intelligence (AI-assisted retrieval, 100% citation correctness), 8-agent LangGraph investigation engine (autonomous tool routing, memory, deterministic HITL escalation for edge cases), and tenant self-service (guardrails, deterministic fallbacks). Full testing strategy, observability, and CI regression gate - 93.6% precision, 87.9% F1 across 1,240 test cases. Zero unsafe outputs shipped.

AgentProbe: LLM Evaluation Platform with CI-Integrated Regression Gate

Quality Controls · Autonomous Safety · Open Source

Python · FastAPI · LangGraph · ChromaDB · CI/CD (GitHub Actions) · Anthropic · OpenAI · Ollama | github.com/venkatalakshmiparimala/AgentProbe

Production LLM quality control platform built end-to-end: 15-endpoint FastAPI service, 5-metric engine (correctness, groundedness, latency, token efficiency, hallucination rate), CI-integrated regression gate that blocks unsafe or degraded outputs before they ship, multi-provider routing for cost optimization. Zero regressions shipped across 1,240+ test cases. The same quality-gate philosophy that belongs in any autonomous campaign system. Open-source.

AI on FHIR: Full-Stack NLP System - Customer-Facing Production

Full-Stack · APIs · ML Integration

Python · Flask · React.js · TypeScript · Docker · BERT · spaCy · LLMs · REST APIs | github.com/venkatalakshmiparimala/AI_on_FHIR

Full-stack customer-facing AI application (Python backend + React.js/TypeScript frontend) translating natural-language queries into FHIR REST API calls via BERT embeddings and LLM inference - reduced retrieval time ~60%; HIPAA-compliant RBAC with authentication, authorization, and full data isolation. End-to-end ownership from architecture to deployment.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering - JNTU Anantapur — 2013-2017